

F Idea Generation Agent: Additional Implementation Details

Seed Idea Generation Due to the max output length limit of the LLM API, we first generate a large number of shorter seed ideas. We keep the seed ideas short so that we can explore more different ideas given the same output token budget. We provide a demonstration example of the seed idea in Appendix G. Then, we perform duplication and expand each remaining seed idea into a full project proposal following our standard template in Appendix B.

Retrieval Augmentation We apply retrieval augmentation to the idea generation prompt in order to increase diversity in the idea generation. To maximize diversity, we apply retrieval augmentation half of the time when generating seed ideas, and we randomly select $k = 10$ papers from the top 20 retrieved papers when applying retrieval augmentation.

Idea Filtering After expanding seed ideas into full project proposals, we did some basic filtering to remove any project proposals that failed the novelty and feasibility checks:

1. Novelty: We use the literature review module to retrieve the top 10 most relevant papers to the generated idea and ask the LLM to compare each of them to the generated idea. The idea will be filtered as long as any one of the retrieved papers is judged as equivalent.
2. Feasibility: The idea will be filtered if it requires extensive manual labor or hardware resources beyond the capacity of a typical academic lab. The idea will also be filtered if it involves any inconsistency in the experimental setups or assumptions. For example, if the idea assumes only black-box API access of the LLMs, then it shouldn't involve experiments that need internal weight access.

This filtered out about 1% of the generated project proposals.

G Demonstration Example: Seed Idea Generation

We present a demonstration example used for seed idea generation. The example is summarized from an existing paper ([Dhuliawala et al., 2023](#)).

Title:

Chain-of-Verification Prompting

Problem:

Generation of plausible yet incorrect factual information, termed hallucination, is an unsolved issue in large language models.

Existing Methods:

A majority of the methods for reducing hallucination can be divided into roughly three categories: training-time correction; generation-time correction; and via augmentation (tool-use).

Motivation:

A key observation is that large language models, when suitably prompted, can both generate and execute a plan of how to verify themselves in order to check their own work, and finally incorporate this analysis into an improved response.

Proposed Method:

Our overall process, which we call Chain-of-Verification (CoVe), thus performs four core steps:

- (1) **Generate Baseline Response:** Given a query, generate the response using the LLM.
- (2) **Plan Verifications:** Given both query and baseline response, generate a list of verification questions that could help to self-analyze if there are any mistakes in the original response.
- (3) **Execute Verifications:** Answer each verification question in turn, and hence check the answer against the original response to check for inconsistencies or mistakes.
- (4) **Generate Final Verified Response:** Given the discovered inconsistencies (if any), generate a revised response incorporating the verification results.

Each of these steps is performed by prompting the same LLM in different ways to obtain the desired response.

Experiment Plan:

Compare with zero-shot prompting, Chain-of-Thought, and few-shot prompting on the MultiSpanQA dataset on closed-book QA and FactScore dataset on generating biographies.

H Generated Seed Ideas and Their Nearest Neighbors

We present several randomly sampled generated seed ideas (see Appendix F for the definition of seed ideas) on the topic of “novel prompting methods that can better quantify uncertainty or calibrate the confidence of large language models”. For each idea, we show the most similar idea (nearest neighbor) based on the embedding similarity, along with the similarity score. In practice, we set a threshold of 0.8 for determining whether two ideas are duplicates.

Idea 1:

Title: Adaptive Precision Boundary Probing

Problem: LLMs often provide uncertainty estimates that are either too coarse-grained or inappropriately precise, failing to adapt to the inherent ambiguity or precision requirements of different queries.

Existing Methods: Existing uncertainty quantification methods typically use fixed precision scales or calibration techniques that don’t adapt to the specific context and precision requirements of each query.

Motivation: Human experts adjust the precision of their uncertainty estimates based on the nature of the question and the available evidence. We can incorporate this adaptive approach to improve LLM uncertainty quantification.

Proposed Method: We introduce Adaptive Precision Boundary Probing (APBP), a dynamic prompting technique that iteratively refines the precision of uncertainty estimates. Given a query, APBP starts with a coarse-grained confidence interval. It then prompts the model to assess whether this interval is appropriately precise given the query’s context and the model’s knowledge. If the model determines that greater precision is warranted, APBP iteratively narrows the interval, prompting the model at each step to justify the increased precision. Conversely, if the model recognizes high ambiguity or limited knowledge, APBP widens the interval. Throughout this process, the model is asked to explicitly reason about the factors influencing the appropriate level of precision, such as the specificity of the query, the reliability of relevant knowledge, and potential sources of ambiguity. The final output is an uncertainty estimate with a precision level tailored to the specific query and the model’s knowledge state.

Experiment Plan: We will evaluate APBP on a diverse set of tasks with varying inherent precision requirements, including numerical estimation, date prediction, and open-ended text generation. We’ll compare APBP against fixed-precision uncertainty estimation methods, measuring both calibration accuracy and the appropriateness of precision levels as judged by human experts.

Nearest Neighbor of Idea 1:

Title: Contextual Confidence Oscillation

Problem: Current methods for quantifying uncertainty in large language models often fail to capture the dynamic nature of confidence across different contexts within a single query.

Existing Methods: Most existing approaches use static confidence scores or calibration techniques that don’t account for intra-query contextual shifts.

Motivation: Human confidence often fluctuates as we process different parts of a complex question or task. By mimicking this oscillation, we can potentially capture a more nuanced and accurate representation of model uncertainty.

Proposed Method: We propose Contextual Confidence Oscillation (CCO), a novel prompting technique that encourages the model to continuously re-evaluate and express its confidence as it processes a query. The prompt is structured as a series of checkpoints, where the model must pause its reasoning, reflect on its current confidence level, and explain any changes since the last checkpoint. This creates a confidence trajectory that can be analyzed for patterns, sudden drops, or gradual increases. Addi-

tionally, we introduce ‘confidence disruptors’ - intentionally ambiguous or challenging sub-queries inserted at various points to test the model’s ability to recognize and express increased uncertainty when appropriate.

Experiment Plan: We will evaluate CCO against standard uncertainty quantification methods on a range of tasks, including multi-step reasoning problems, ambiguous queries, and long-form text analysis. We’ll measure not just overall accuracy of uncertainty estimates, but also the correlation between confidence oscillations and human-annotated difficulty levels of different parts of each query. We’ll also analyze how well the model’s expressed confidence trajectory aligns with its actual performance across different segments of complex tasks.

Similarity: 0.70

Idea 2:

Title: Quantum Superposition Confidence Prompting

Problem: Current LLMs struggle to accurately quantify uncertainty across multiple possible answers, often defaulting to overconfidence in a single response.

Existing Methods: Existing approaches typically involve single-path reasoning or limited branching, failing to capture the full spectrum of uncertainty.

Motivation: Inspired by quantum mechanics, where particles can exist in multiple states simultaneously, we propose a method that allows LLMs to consider multiple answer possibilities concurrently.

Proposed Method: We introduce Quantum Superposition Confidence Prompting (QSCP), where the LLM is instructed to generate multiple potential answers simultaneously, assigning confidence scores to each. The prompt encourages the model to ‘exist in multiple states,’ exploring contradictory answers and their implications concurrently. For example: ‘Imagine you are in a quantum superposition of multiple expert personas. Each persona will provide an answer to the following question, along with a confidence score (0-100%). Ensure the personas explore contradictory viewpoints. Question: [INSERT QUESTION]’. The LLM then generates responses from multiple personas, each with its own confidence score. The final uncertainty is derived from the distribution of these scores, providing a more nuanced understanding of the model’s confidence across possible answers.

Experiment Plan: Compare QSCP against standard prompting, chain-of-thought, and other uncertainty quantification methods on diverse question-answering datasets. Evaluate using metrics such as calibration error, Brier score, and a novel ‘quantum uncertainty score’ that measures the spread and coherence of the generated answer superposition.

Nearest Neighbor of Idea 2:

Title: Quantum Superposition Prompting

Problem: Traditional methods for uncertainty quantification in large language models often fail to capture the full range of possible interpretations and outcomes, especially for queries with inherent ambiguity or multiple valid perspectives.

Existing Methods: Current approaches typically focus on generating a single response with an associated confidence score, or at best, a small set of discrete alternatives.

Motivation: Drawing inspiration from the principle of superposition in quantum mechanics, we propose a method to represent and reason about multiple possible outcomes simultaneously, providing a richer and more nuanced uncertainty quantification.

Proposed Method: We present Quantum Superposition Prompting (QSP), a novel framework for exploring and quantifying uncertainty in language model outputs. QSP begins by prompting the model to generate a ‘superposition’ of possible interpretations or approaches to the given query. Each element in this superposition is assigned a complex amplitude, representing both its probability

and its relationship to other elements. The model is then guided through a series of ‘measurement’ prompts, designed to collapse this superposition along different bases of interpretation. These measurements yield probability distributions over outcomes, capturing different facets of uncertainty. QSP employs techniques inspired by quantum computing, such as interference and entanglement, to model how different interpretations interact and influence each other. The final uncertainty quantification is derived from the full set of measurements, providing a multi-dimensional representation of the model’s uncertainty that captures ambiguity, conflicting evidence, and the interdependence of different interpretations.

Experiment Plan: We will evaluate QSP on tasks that inherently involve multiple valid perspectives or ambiguous interpretations, such as ethical dilemmas, creative writing prompts, and open-ended analytical questions. Metrics will include the diversity and coherence of generated superpositions, the ability to capture human-judged ambiguities, and improvements in uncertainty calibration compared to classical methods.

Similarity: 0.77

Idea 3:

Title: Fractal Uncertainty Decomposition

Problem: LLMs often provide overly simplistic uncertainty estimates that fail to capture the hierarchical and nested nature of uncertainty in complex knowledge domains.

Existing Methods: Current uncertainty quantification methods typically produce flat, single-dimensional confidence scores that don’t reflect the multi-layered structure of knowledge and uncertainty.

Motivation: By recursively decomposing a query into sub-components and assessing uncertainty at multiple levels of granularity, we can construct a more comprehensive and structurally informed uncertainty estimate.

Proposed Method: We introduce Fractal Uncertainty Decomposition (FUD), a prompting technique that recursively breaks down a query into a hierarchical structure of sub-queries, assessing uncertainty at each level. Given an initial query, FUD prompts the model to identify key sub-components or aspects of the question. For each sub-component, the model provides an answer and a confidence estimate. If the confidence for a sub-component is below a certain threshold, FUD recursively applies the same decomposition process to that sub-component. This continues until either a maximum depth is reached or all sub-components have high confidence. The resulting structure is a tree of nested confidence estimates. FUD then aggregates these estimates bottom-up, using a combination of statistical methods and prompted meta-analysis by the model. The final output is both an overall uncertainty estimate and a detailed map of the uncertainty structure, showing how confidence varies across different aspects and levels of the query.

Experiment Plan: We will evaluate FUD on complex, multi-faceted tasks such as scientific explanation, historical analysis, and technical troubleshooting. We will compare its performance to flat confidence estimation methods and other hierarchical approaches. Evaluation metrics will include traditional calibration measures, as well as new metrics designed to assess the quality and informativeness of the uncertainty decomposition. We will also conduct case studies to demonstrate how FUD can provide more actionable and interpretable uncertainty information in real-world scenarios.

Nearest Neighbor of Idea 3:

Title: Semantic Fractal Decomposition

Problem: Current uncertainty quantification methods for large language models often fail to capture the hierarchical and self-similar nature of conceptual understanding, leading to inconsistent confi-